

Breaking Bad: Component-Wise Parent Deletion for Score-Based Causal Discovery

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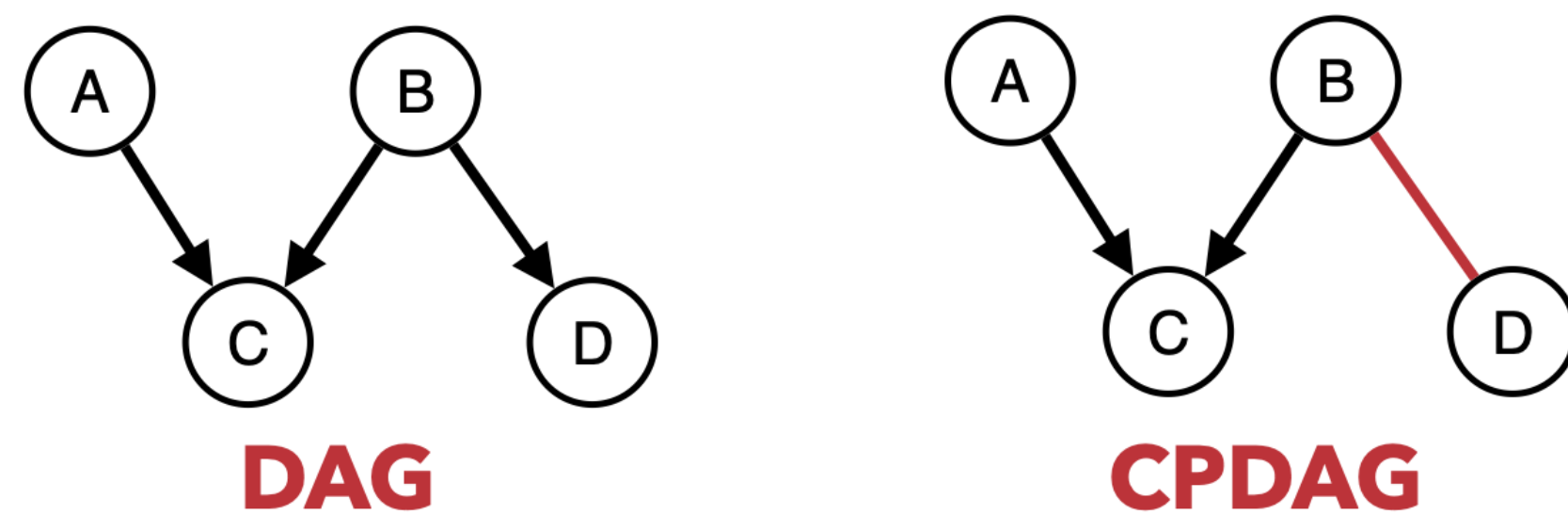
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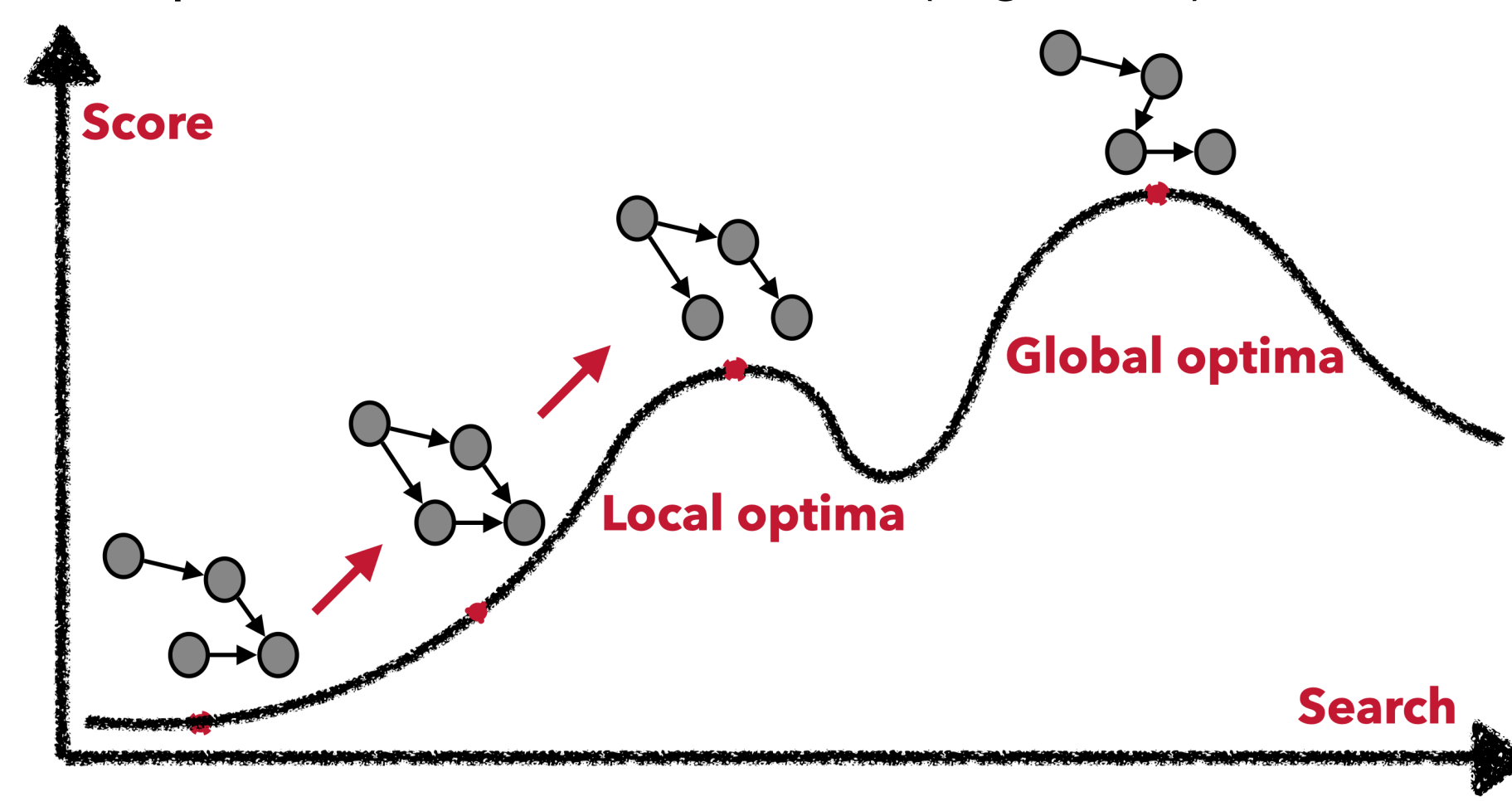
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Background

- **Markov Equivalence Class (MEC)** DAGs sharing the same conditional independencies form a Markov Equivalence Class, represented by a CPDAG.

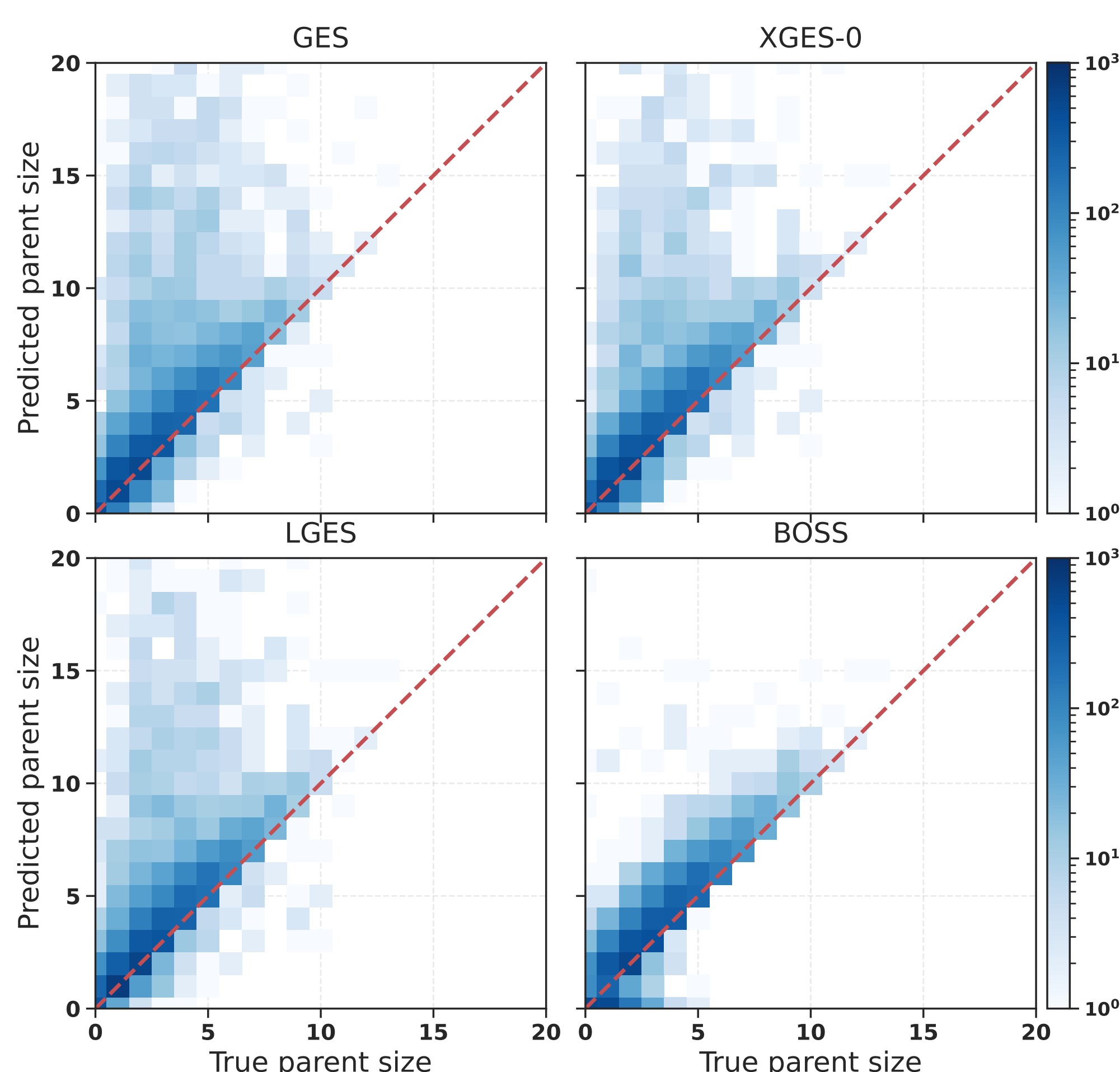


- **Score-based causal discovery** learns a causal graph from observational data by searching for the structure that optimizes a score function (e.g., BIC).



Motivation

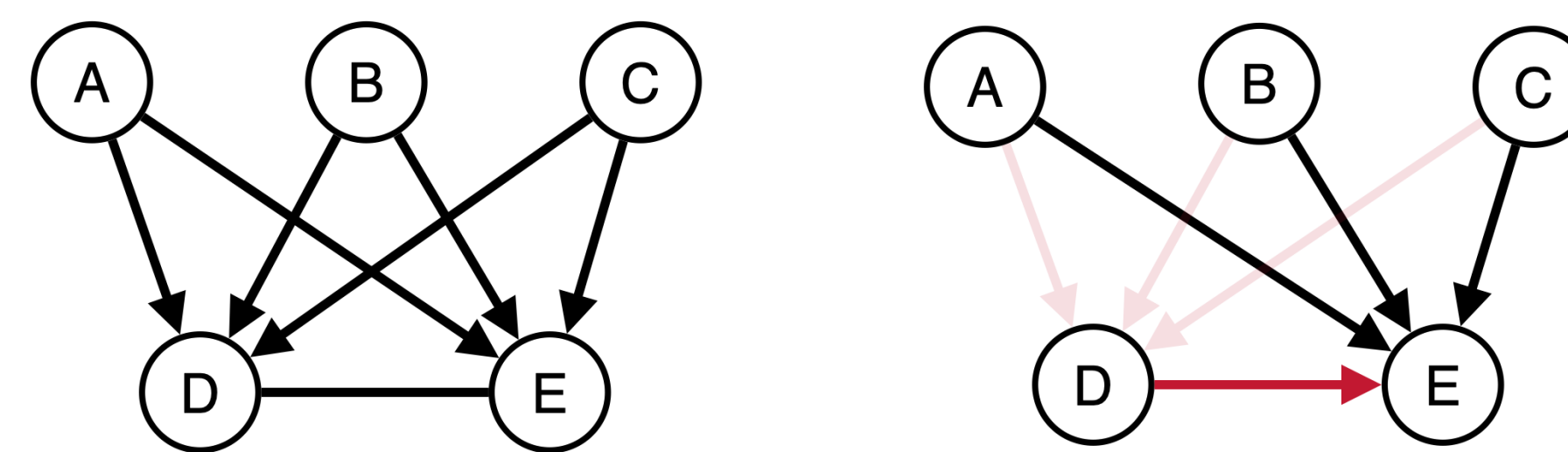
- **Scalability and search challenges** Score-based search struggles in high-dimensional and dense settings. In finite samples, it often becomes trapped in local optima and fails to recover the true MEC.



- **Parent set over-estimation** Score-based methods tend to introduce excessive parent variables, producing overly dense structures.

Parent deletion

- **Parent deletion** Remove all incoming edges of a node and reorient selected neighboring edges while preserving CPDAG validity.



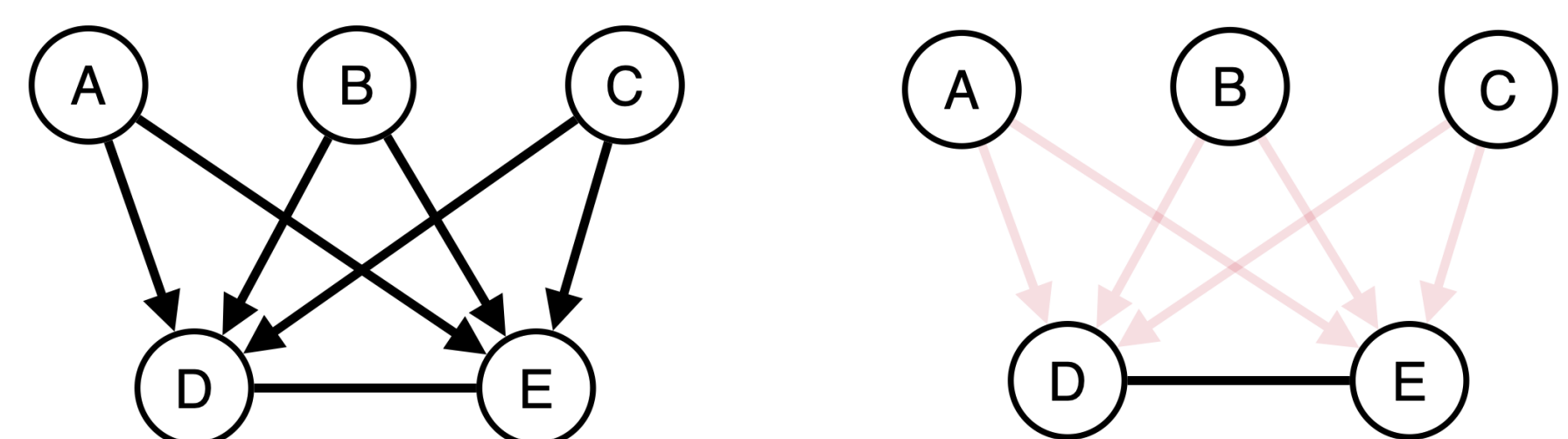
“Merely deleting many parent edges leads to unintended side effects.”

- **Collateral parental growth**

1. Nodes connected through undirected edges share the same parent set.
2. Parent deletion may increase neighboring parent sets.

Component-wise parent deletion

- **Component-wise parent deletion** Delete parents for an entire undirected component simultaneously.



- **Validity** component-wise parent deletion is always valid in a CPDAG and requires no additional validity checking.
- **Structural remedy** By deleting parents of the entire component, the shared-parent structure is removed simultaneously, preventing collateral parental growth.

“Simple yet powerful operator theoretically sound, effectively alleviating scalability issues.”

Iterated Local Search (ILS)

- **Iterated Local Search** Escape local optima by repeatedly perturbing and re-optimizing the current solution.

Method	Perturbation
-D	Single-edge deletion
-DP	Component-wise parent deletion
DP+	Scheduling
	Component-wise → Node-wise → Edge-wise

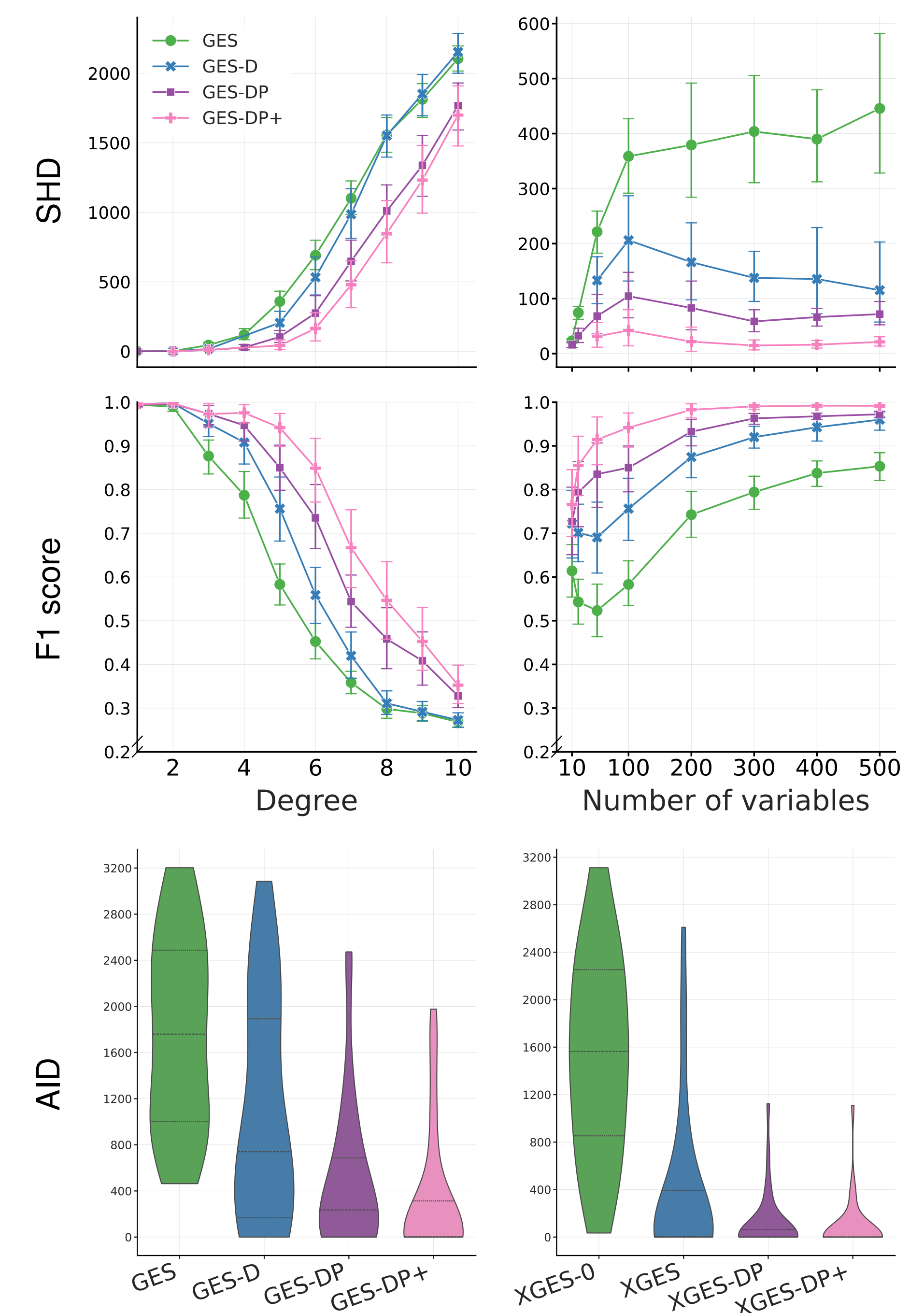
- **Plug-in** Our perturbation can be seamlessly incorporated into existing score-based search algorithms.

Experimental setup

- **Baselines** GES, OPS, XGES, LGES, and BOSS.
- **Evaluation metrics** SHD (↓): structural distance, F1 (↑): edge recovery accuracy, AID (↓): causal effect distance, Runtime (↓): computational cost.
- **Evaluation protocol** For each score-based search algorithm, we compare different perturbation strategies.

Results

- **Synthetic datasets** Erdős–Rényi, Scale-Free network



- **Biological networks** ARTH150, ECOLI70, MAGIC-IRRI, MAGIC-NIAB.

